Intro to Deep Learning

A Beginner's Journey into Artificial Neural Networks

What is Deep Learning?

The core technology behind modern AI breakthroughs.

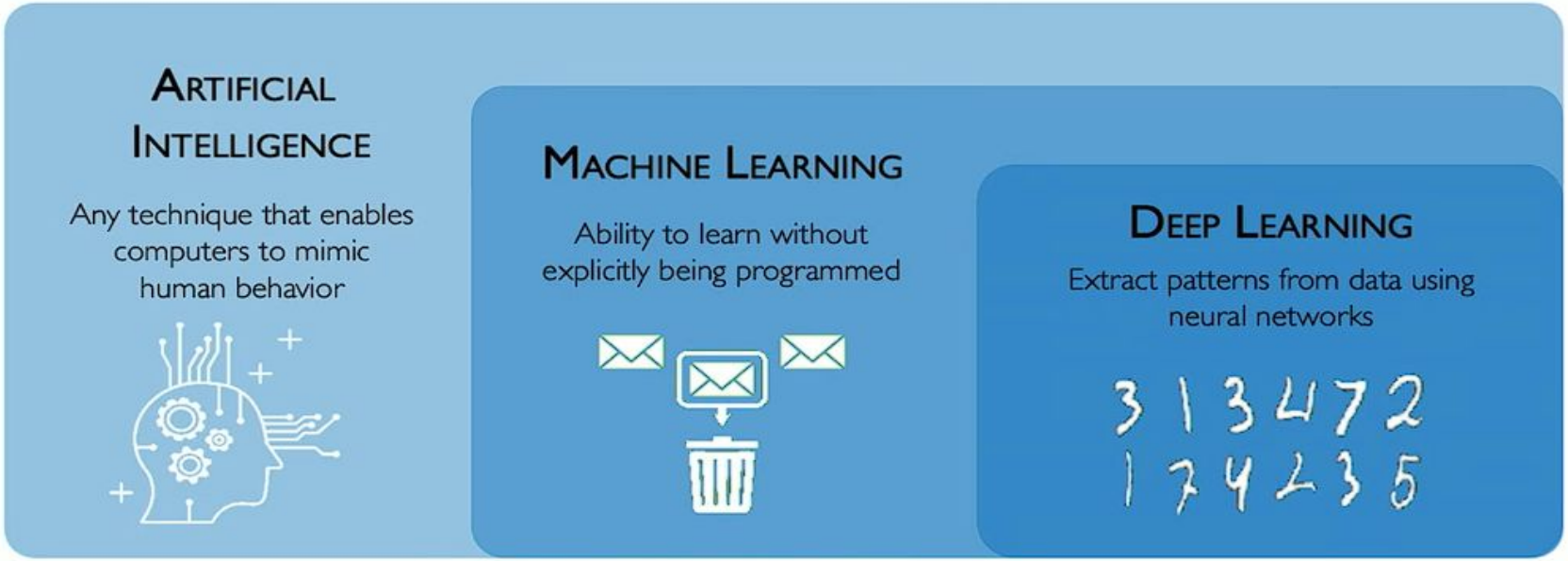
Defining the Concept

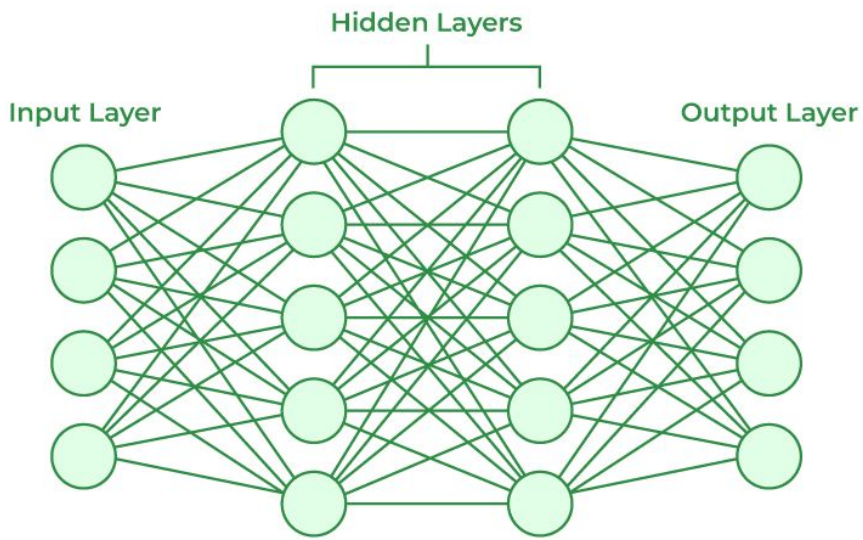
A Subset of AI

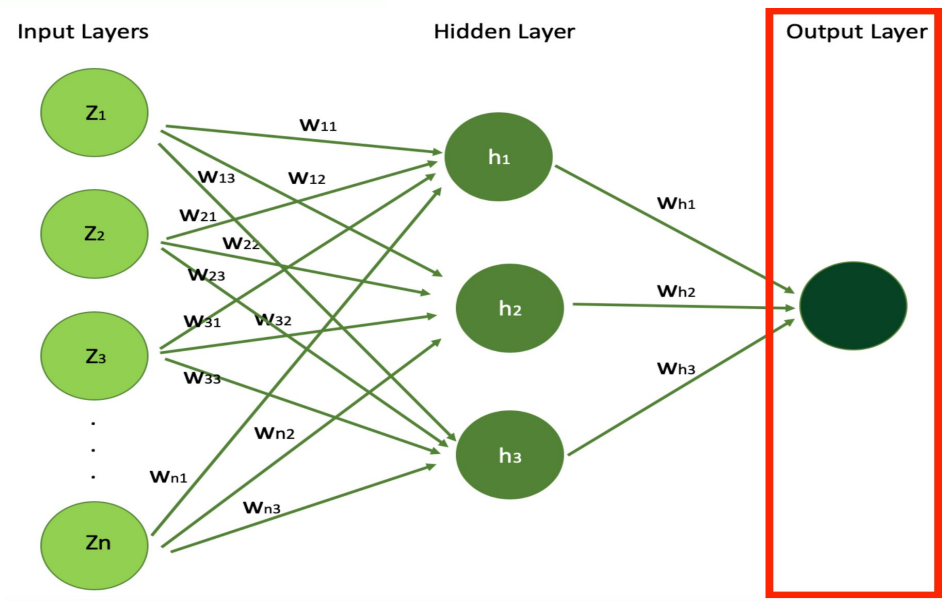
Deep Learning is a specific subfield of Machine Learning. While all Deep Learning is AI, not all AI uses Deep Learning techniques.

Multi-Layered Learning

It utilizes "Deep" neural networks—models with many layers of processing that mimic how information flows through biology.


ARTIFICIAL INTELLIGENCE
Any technique that enables computers to mimic human behavior
MACHINE LEARNING
Ability to learn without explicitly being programmed
DEEP LEARNING
Extract patterns from data using neural networks


Hidden Layers
Input Layer
Output Layer


Input Layers
Hidden Layer
Output Layer
Z1
Z2
Z3
Zn
h1
h2
h3
W11
W12
W13
W21
W22
W23
W31
W32
W33
Wn1
Wn2
Wn3
Wh1
Wh2
Wh3

The Neural Network

Understanding the architecture of a machine brain.

History of Neural Networks

1943

McCulloch & Pitts model the first mathematical neuron.

1958

Frank Rosenblatt creates the "Perceptron" for vision.

1986

The discovery of Backpropagation allows deep training.

2012

AlexNet wins ImageNet, sparking the modern DL era.

Early Precursors and Mathematics

- **1960:** Henry J. Kelley developed a continuous-time version of control theory optimization that served as an early precursor.
- **1962:** Frank Rosenblatt used the term "back-propagating error correction," though he did not implement it for multi-layer models.
- **1970:** Seppo Linnainmaa published the first computer implementation of the reverse mode of automatic differentiation (the core math behind backpropagation) in his master's thesis. Reddit · r/MachineLear... +3

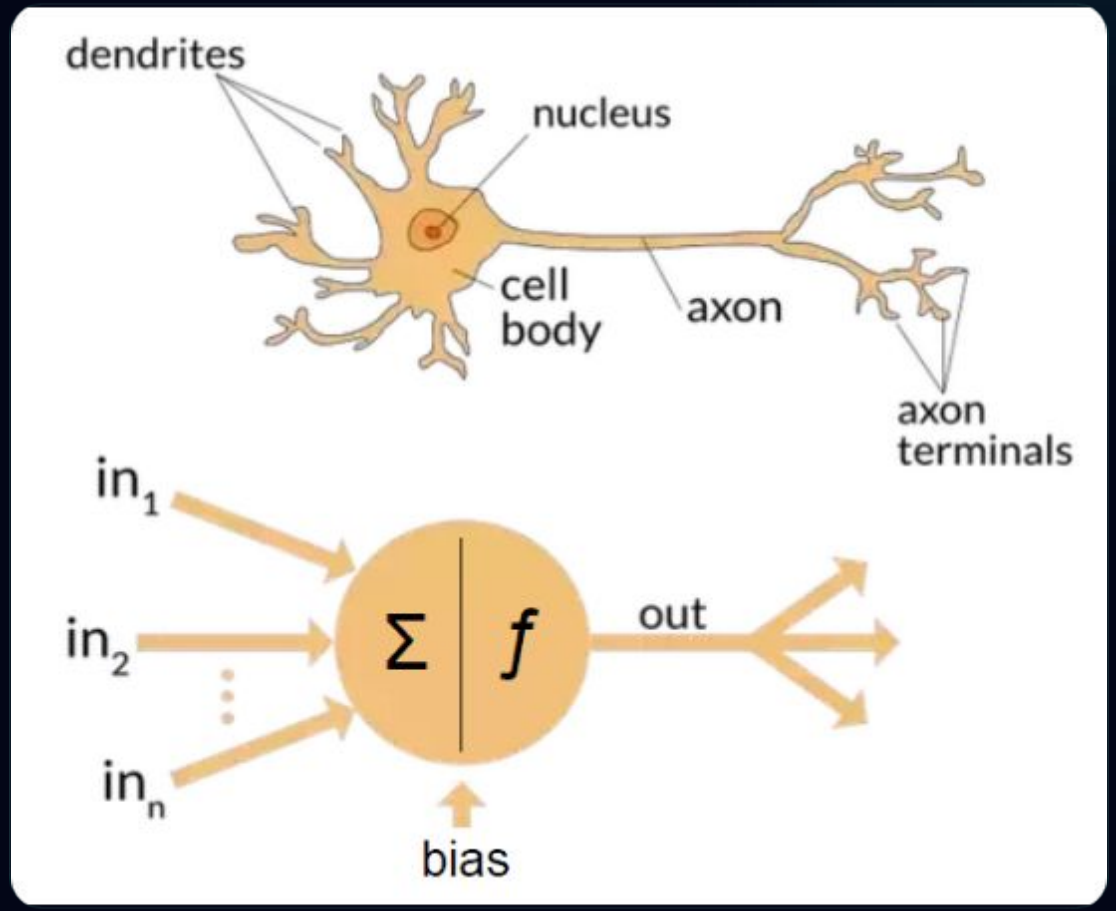
Application to Neural Networks

- **1974:** Paul Werbos was the first to propose applying this backward-flowing error derivative process specifically to artificial neural networks in his PhD thesis. YouTube · CodeEmporiu... +1
- **1982:** Paul Werbos further detailed the application of backpropagation to multi-layer perceptrons. Wikipedia +1
- **1986:** David Rumelhart, Geoffrey Hinton, and Ronald Williams published their landmark paper, "Learning representations by back-propagating errors," which proved the method worked efficiently and made it famous worldwide. Michael Brenndoerfer +2

The Brain Connection

Biological Inspiration

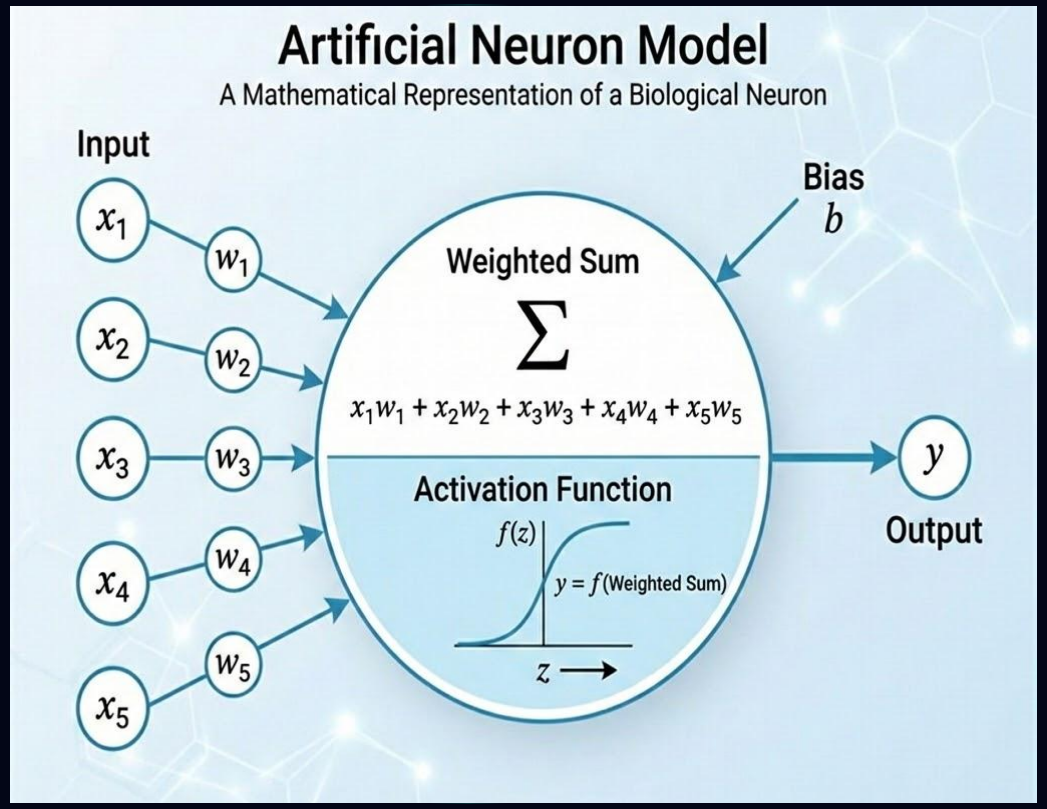
Deep Learning was inspired by the human brain's web of neurons. Just as biological synapses strengthen through learning, AI "weights" are adjusted to improve accuracy. While the brain uses electro-chemical signals, artificial networks use mathematical operations to process information across millions of connections.



Network Architecture

- **Input Layer:** The gateway that accepts raw data such as pixel values, audio signals, or text tokens.
- **Hidden Layers:** Multiple layers where the actual "learning" happens by extracting increasingly complex features.
- **Output Layer:** The final stage that provides the prediction, such as "Cat" vs. "Dog" or a price estimate.

The Artificial Neuron



The Unit of Calculation

An artificial neuron (node) receives input signals, multiplies them by **Weights**, adds a **Bias**, and passes the result through an **Activation Function** to determine the output.

Why Deep Learning?

The shift from traditional Machine Learning to DL.

The DL Advantage



Auto-Extraction

DL automatically finds the best features in data, removing the need for manual "feature engineering."



Unstructured Data

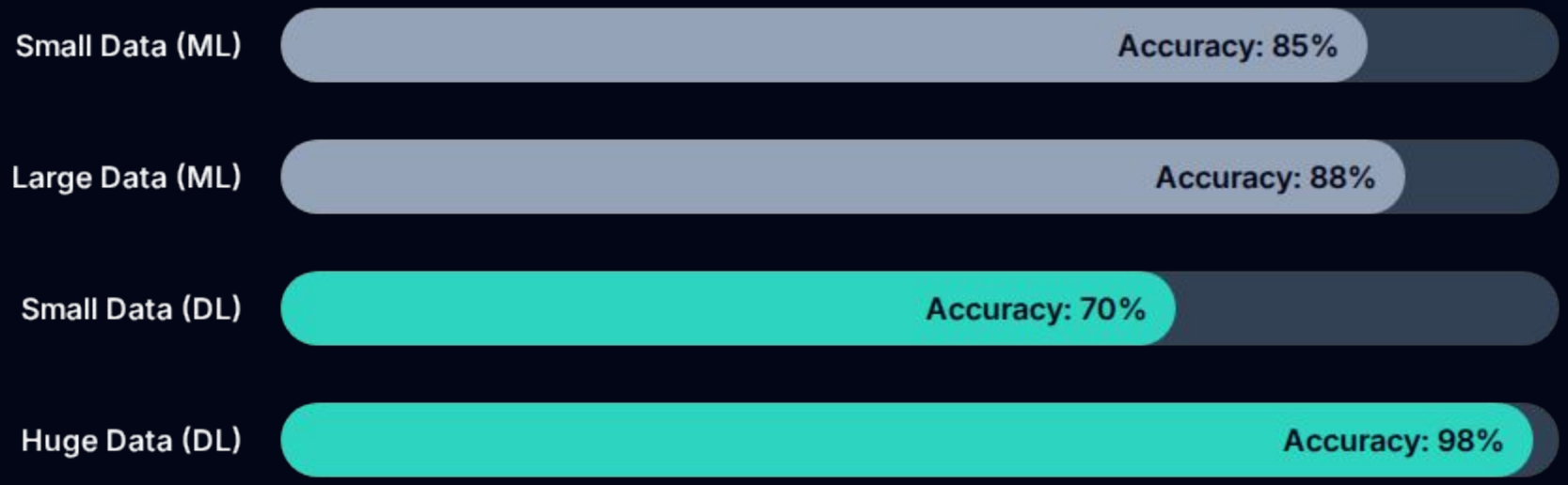
It excels at processing "messy" data like high-res images, raw audio, and natural language.



High Scalability

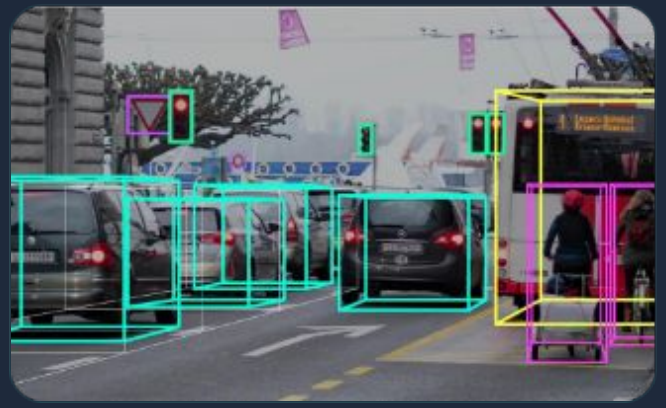
While traditional ML performance plateaus, DL accuracy continues to improve as you add more data.

Performance vs. Data Volume



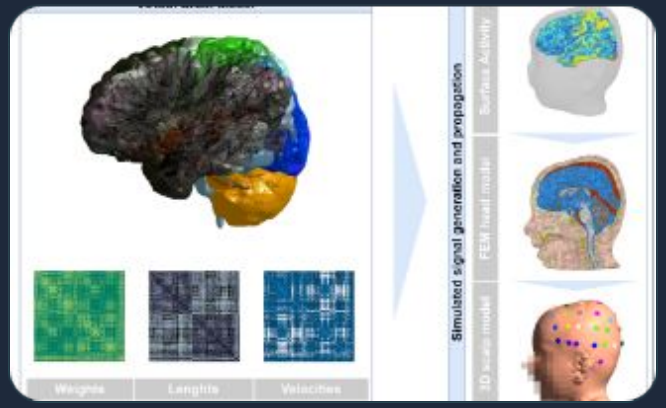
Deep Learning models require vast datasets to surpass traditional algorithms, but their potential ceiling is much higher.

Real-World Use Cases



Self-Driving

Identifying objects, lanes, and signs in real-time.



Medical Imaging

Detecting diseases in X-rays and MRI scans with high precision.



NLP & Chatbots

Powering tools like ChatGPT and Siri to understand human speech.

Questions?

Deep Learning is just beginning to transform our world.

Thank you for your attention!